

What is Flutter? (Big Picture)

1. Introduction

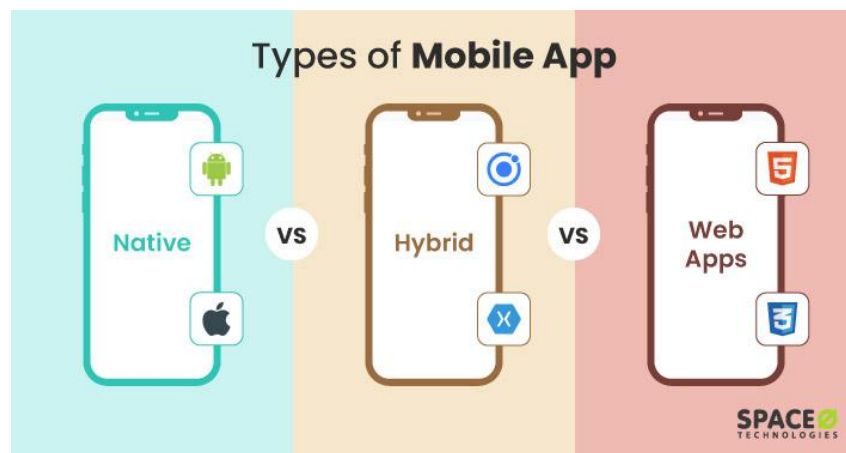
Flutter is an **open-source UI software development kit (SDK)** created by **Google**. It is used to build **natively compiled applications** for **mobile (Android & iOS)**, **web**, and **desktop** from a **single codebase**.

In simple words:

Flutter allows developers to write one code and run it on multiple platforms with high performance and beautiful UI.

2. Types of Applications (Before Flutter)

Before learning Flutter, students should understand the **different types of applications** that exist today.



2.1 Native Apps

Native apps are built specifically for a single platform using its official language and tools.

- Android: Java / Kotlin
- iOS: Swift / Objective-C

Characteristics:

- Best performance
- Full access to device features
- Separate codebase for each platform

Examples: WhatsApp, Instagram (native core)

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2.2 Web Apps

Web apps run inside a web browser and are built using:

- HTML
- CSS
- JavaScript

They do not need installation from an app store.

Characteristics:

- Runs in browser
- Easy to update
- Limited access to device hardware

Examples: Gmail, Google Docs

2.3 Hybrid Apps

Hybrid apps are web apps wrapped inside a native container.

They use:

- HTML
- CSS
- JavaScript

And run inside a **WebView**.

Characteristics:

- Single codebase
- Slower performance than native
- UI depends on browser rendering

Examples: Early versions of Ionic / Cordova apps

3. Why Flutter Was Created

Before Flutter, developers usually faced these problems:

- Separate codebases for Android and iOS
- High development and maintenance cost

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- Inconsistent UI across platforms
- Slower UI performance in hybrid frameworks

Flutter was created to solve these issues by:

- Providing a **single codebase**
- Delivering **near-native performance**
- Allowing **full control over UI design**
- Enabling **fast development with Hot Reload**

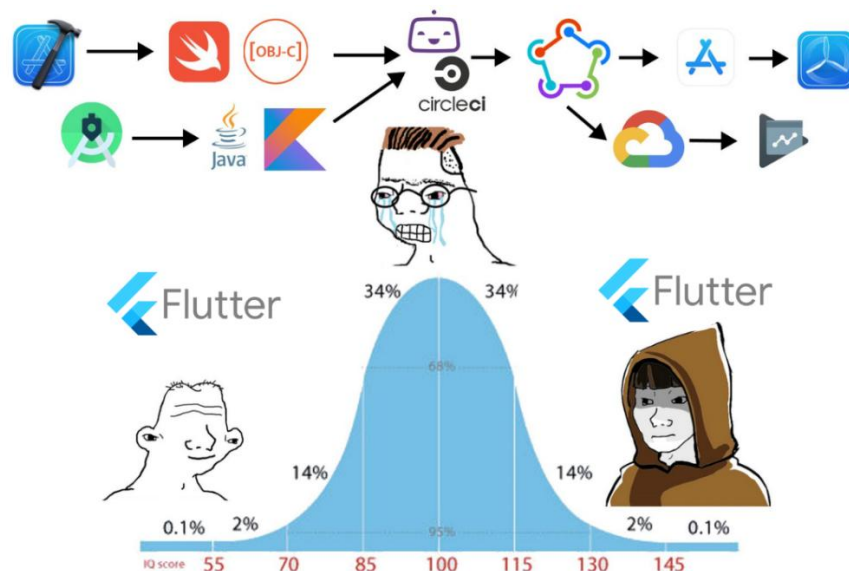
3. What Makes Flutter Different

3.1 Single Codebase

With Flutter, you write your application **once** using Dart, and it works on:

- Android
- iOS
- Web
- Windows
- macOS
- Linux

This saves time, cost, and effort.



3.2 Everything is a Widget

In Flutter, **everything is a widget**:

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- Text
- Button
- Image
- Padding
- Screen layout

Widgets are used to describe **how the UI should look**, not how to draw it.

This makes UI building:

- Consistent
 - Reusable
 - Easy to manage
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3.3 Flutter Does NOT Use Native UI Components

Unlike other frameworks, Flutter:

- Does NOT use Android XML widgets
- Does NOT use iOS UIKit widgets

Instead, Flutter draws everything using its **own rendering engine**.

This results in:

- Same UI on all platforms
 - No platform-specific UI bugs
 - Better performance control
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4. Dart Language (Flutter's Backbone)

Flutter applications are written in **Dart**.

Why Dart?

- Fast execution
- Supports both **JIT** (development) and **AOT** (production)
- Optimized for UI development

Dart enables features like:

- Hot Reload
- Hot ReStart
- Smooth animations

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- Predictable performance
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5. Flutter Architecture (High-Level View)

Flutter is built using a layered architecture:

1. Flutter Framework (Dart)

- Widgets
- State management
- UI logic
- Gestures

2. Flutter Engine (C++)

- Rendering (Skia)
- Text layout
- Image handling
- Dart runtime

3. Platform Embedder

- Connects Flutter to Android, iOS, Web, Desktop
- Handles lifecycle, input, and native APIs

This architecture makes Flutter **fast and reliable**.

6. Hot Reload (Developer Superpower)

Hot Reload allows developers to:

- See UI changes instantly
- Modify code without restarting the app
- Experiment with UI quickly

This drastically improves:

- Learning speed
 - Development productivity
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7. What Can You Build with Flutter

Using Flutter, you can build:

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- Mobile apps (Android & iOS)
- Web applications
- Desktop applications
- MVPs and startups
- Production-level enterprise apps

Popular app categories:

- E-commerce
- Social media
- News & blogs
- Finance & fintech
- Education apps

8. Flutter vs Traditional Development

Feature	Flutter	Native Development
Codebase	Single	Separate
UI Consistency	Same everywhere	Platform dependent
Performance	Near-native	Native
Development Speed	Very fast	Slower
Hot Reload	Yes	No

9. Who Should Learn Flutter

Flutter is ideal for:

- Students learning app development
 - Beginners entering mobile development
 - Web developers moving to mobile
 - Startups building fast MVPs
 - Companies targeting multiple platforms
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10. Summary (Big Picture)

- Flutter is a **cross-platform UI framework**
- Uses **one codebase** for many platforms
- Written in **Dart**
- Uses its **own rendering engine**
- Focuses on **performance, UI consistency, and developer productivity**

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Flutter is not just a framework — it is a **complete ecosystem for modern app development**.

11. What to Learn Next

After understanding what Flutter is, the next topics should be:

1. Dart basics
2. Flutter project structure
3. Widgets (Stateless vs Stateful)
4. Layout system
5. State management basics

This creates a strong foundation for Flutter development.